



Aishu R 251001007-2025-IT ▾

A2**Started on** Monday, 24 August 2026, 12:26 PM**State** Finished**Completed on** Monday, 24 August 2026, 12:27 PM**Time taken** 1 min 13 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int count(int a[], int left, int right, int x)
4  {
5      int i, c = 0;
6
7      for (i = left; i <= right; i++)
8      {
9          if (a[i] == x)
10             c++;
11      }
12
13      return c;
14  }
15
16  int majorityElement(int a[], int left, int right)
17  {
18      int mid, l, r, lc, rc;
19
20      if (left == right)
21          return a[left];
22
23      mid = (left + right) / 2;
24
25      l = majorityElement(a, left, mid);
26      r = majorityElement(a, mid + 1, right);
27

```

```

28     if (l == r)
29         return l;
30
31     lc = count(a, left, right, l);
32     rc = count(a, left, right, r);
33
34     if (lc > rc)
35         return l;
36     else
37         return r;
38 }
39
40 int main()
41 {
42     int n, i;
43     int a[50000];
44
45     scanf("%d", &n);
46
47     for (i = 0; i < n; i++)
48         scanf("%d", &a[i]);
49
50     printf("%d", majorityElement(a, 0, n - 1));
51
52     return 0;

```

	Input	Expected	Got	
✓	3	3	3	✓
	3 2 3			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

